Task 1: Black-Box Testing for Calculator

Sample output

Input:

Enter first number: 10

Enter second number: 5

Output:

Addition: 15.0

Subtraction: 5.0

Division: 2.0

Task 2: White-Box Testing for a Function with If-Else and Loops

Sample output

-2 is not positive

Test 1 - Positive count: 2

-1 is not positive

0 is not positive

Test 2 - Positive count: 0

Test 3 - Positive count: 0

Task 3: SimpleMath – JUnit Testing

Expected output

JUnit version 4.x

.....

Time: 0.01

OK (5 tests)

Task 4: Test String Utilities – Reverse String and Palindrome Check

Expected output

JUnit version 4.x

.....

Time: 0.01

OK (5 tests)

Task 5: Test Divide Method Using assertThrows()

Expected output

Test run finished

4 tests found

4 tests successful

0 tests failed

Task 6: Validate Sorting Function Correctness

Expected output

Test run finished

6 tests found

6 tests successful

0 tests failed

Task 7: List Management with Lifecycle

Expected output

List initialized

Elements added: [Java, Python]

List cleared

1 test successful

0 tests failed

Task 8: Test Lifecycle Execution Order

Expected output

1. Before All

2. Before Each

3. Test One

4. After Each

2. Before Each

3. Test Two

4. After Each

5. After All

Task 9: Test Empty Input, Null Input, and Single Element

Expected output

Tests found: 4

Tests successful: 4

Tests failed: 0

Task 10: Design Test Cases for a Login System

Expected output

Tests found: 10

Tests successful: 10

Tests failed: 0